Ethanol Blending in Gasoline - Colombia

August 2025

U.S. GRAINS & BIOPRODUCTS COUNCIL

20 F St. NW, Suite 900 \ Washington, DC 20001
202-789-0789 \ www.grains.org

FARO90

www.faro90.com
+52 55 6122 2255

Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.

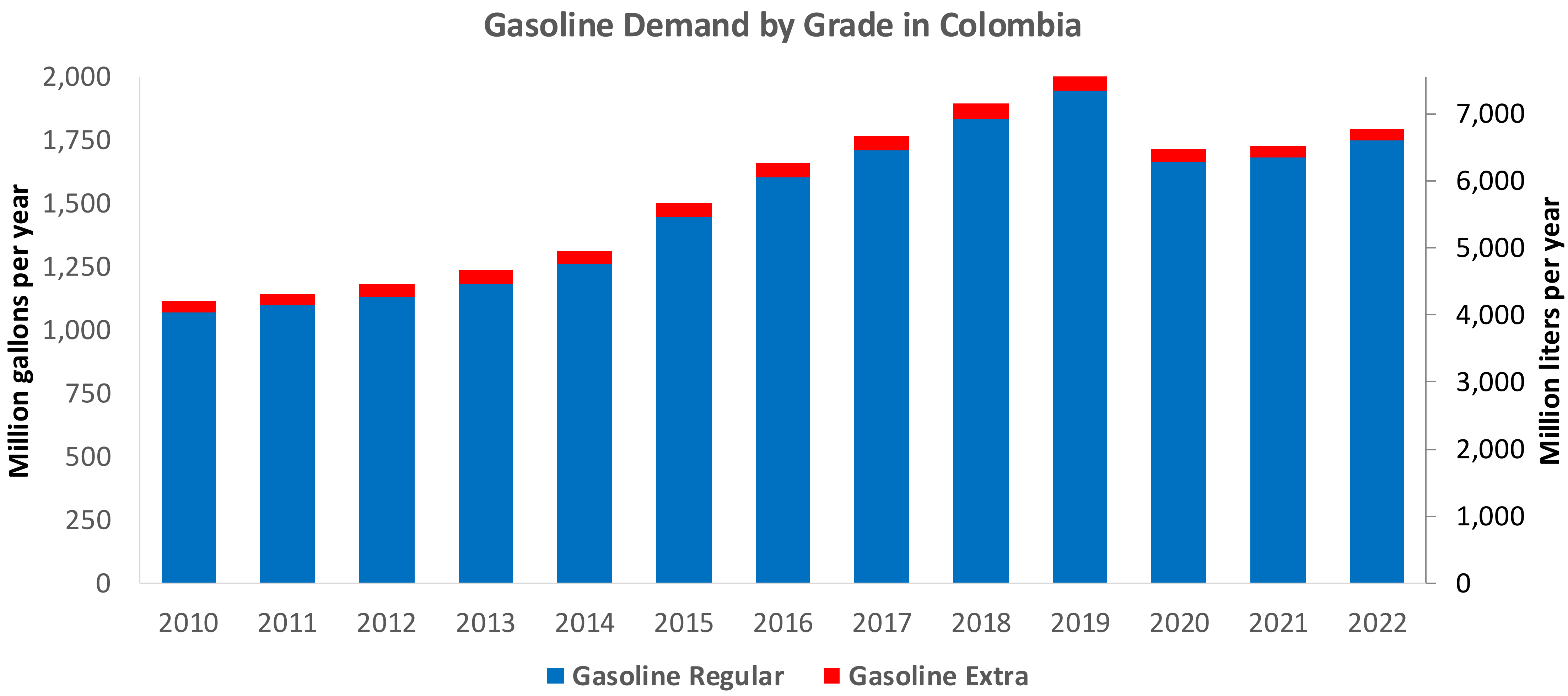
Ethanol Blending in Ethanol - Colombia

In 2024, gasoline consumption in Colombia was 2,023 million gallons (7,658 million liters). Market share was 3% for Extra gasoline E10 (RON 97 – AKI 94) and 97% for Regular gasoline (regular) E10 (RON 89 – AKI 84). Local production supplies 65% of total demand. United States is the main source of gasoline imports.

Ethanol blends are obligatory in cities with more than 500,000 habitants. Resolution 40444 of 2023 authorizes the blending of up to 10% ethanol in the country and allows for lower percentages in particular cases. E10 is the predominant grade in the country except on the border with Venezuela. The ethanol supply comes from imports from the United States and from local production.

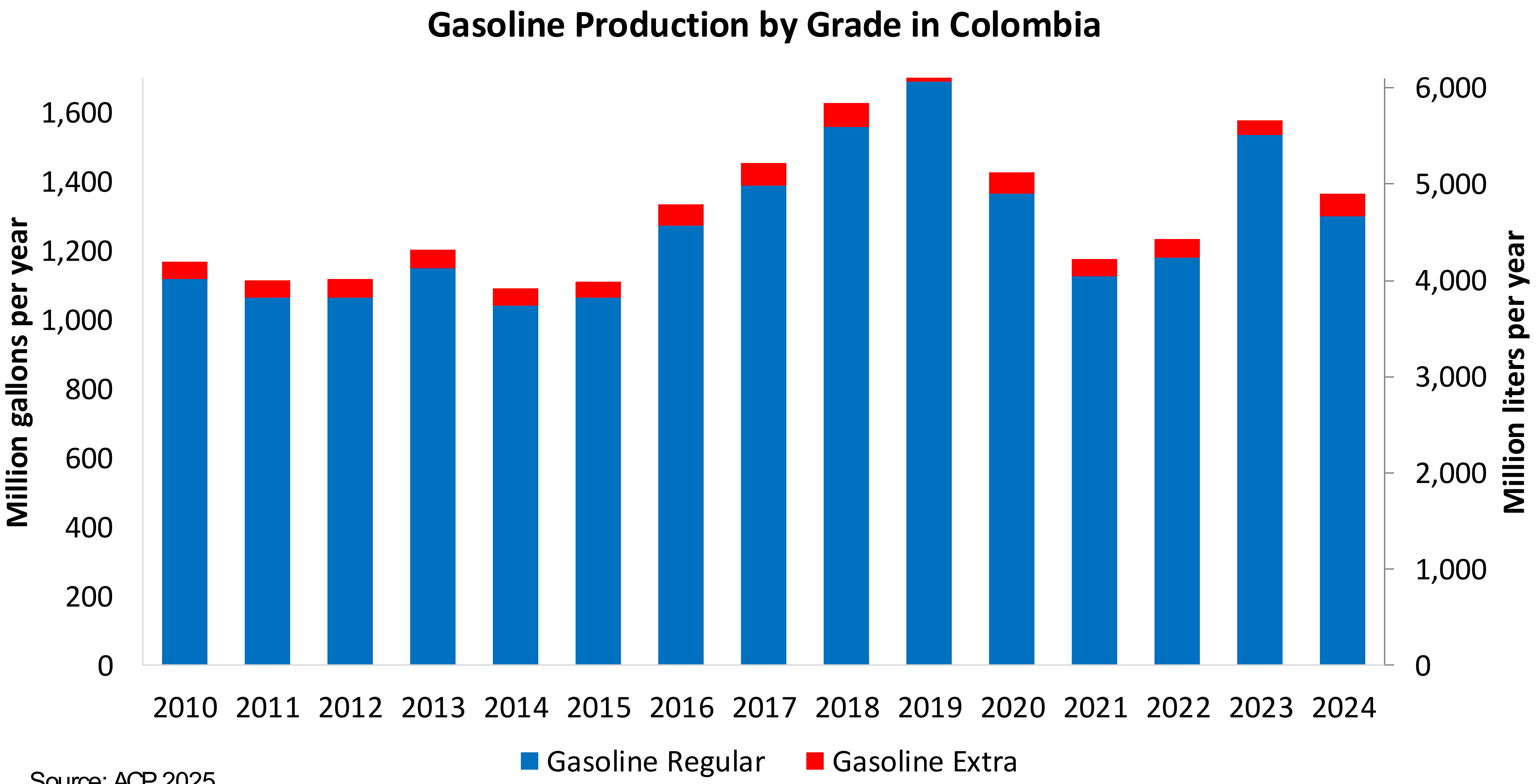
Source: ACP, 2024

Gasoline Demand in Colombia



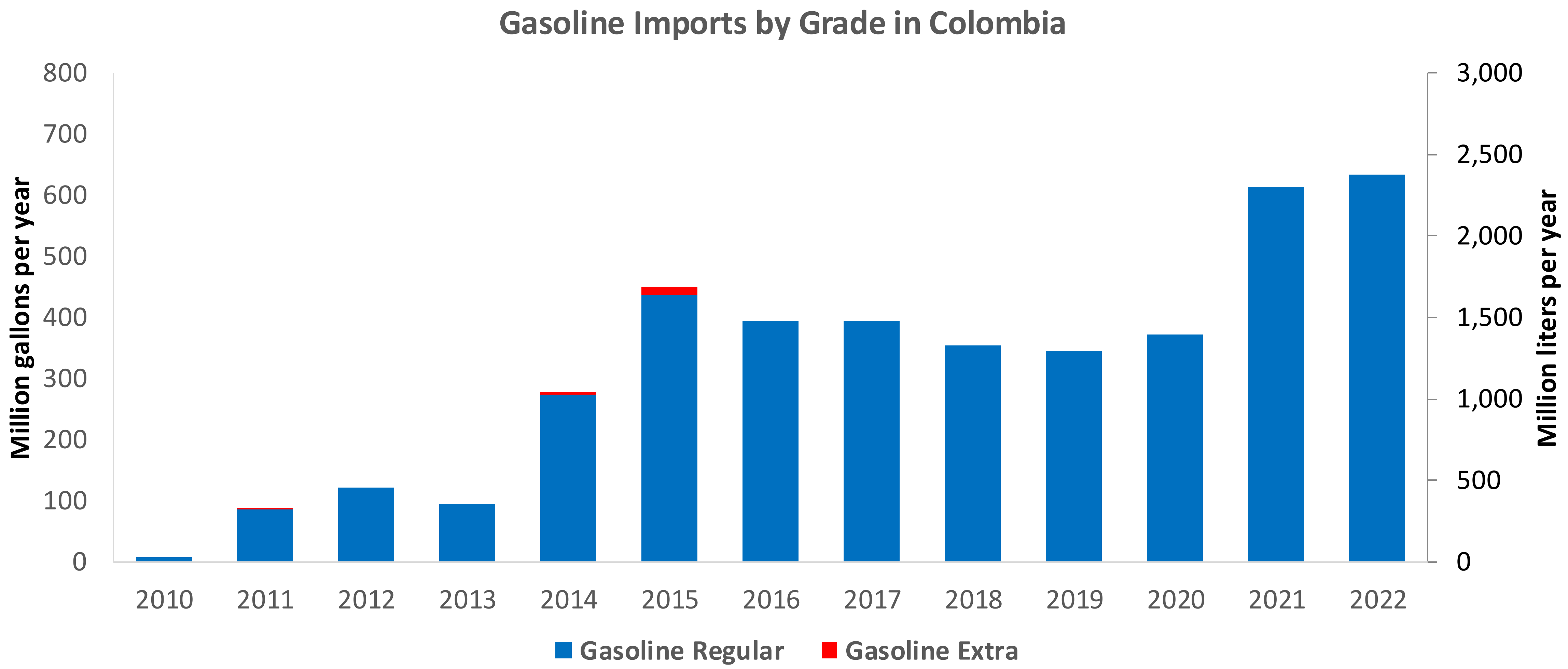
Source: ACP, 2025

Gasoline Production in Colombia



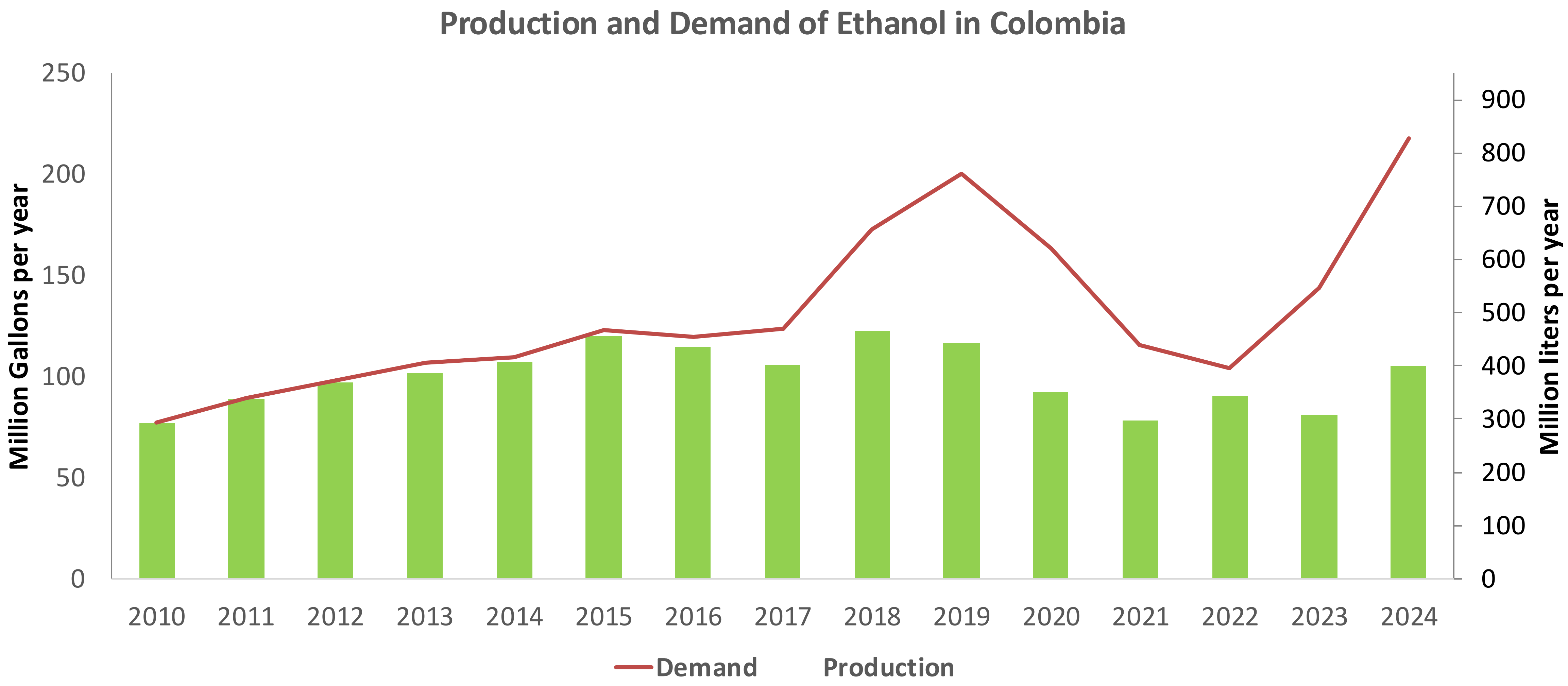
Source: ACP, 2025

Gasoline Imports in Colombia



Source: ACP, 2025

Ethanol Balance in Colombia



Source: Asocaña - Balance Azucarero Colombiano, 2024; EIA, 2024

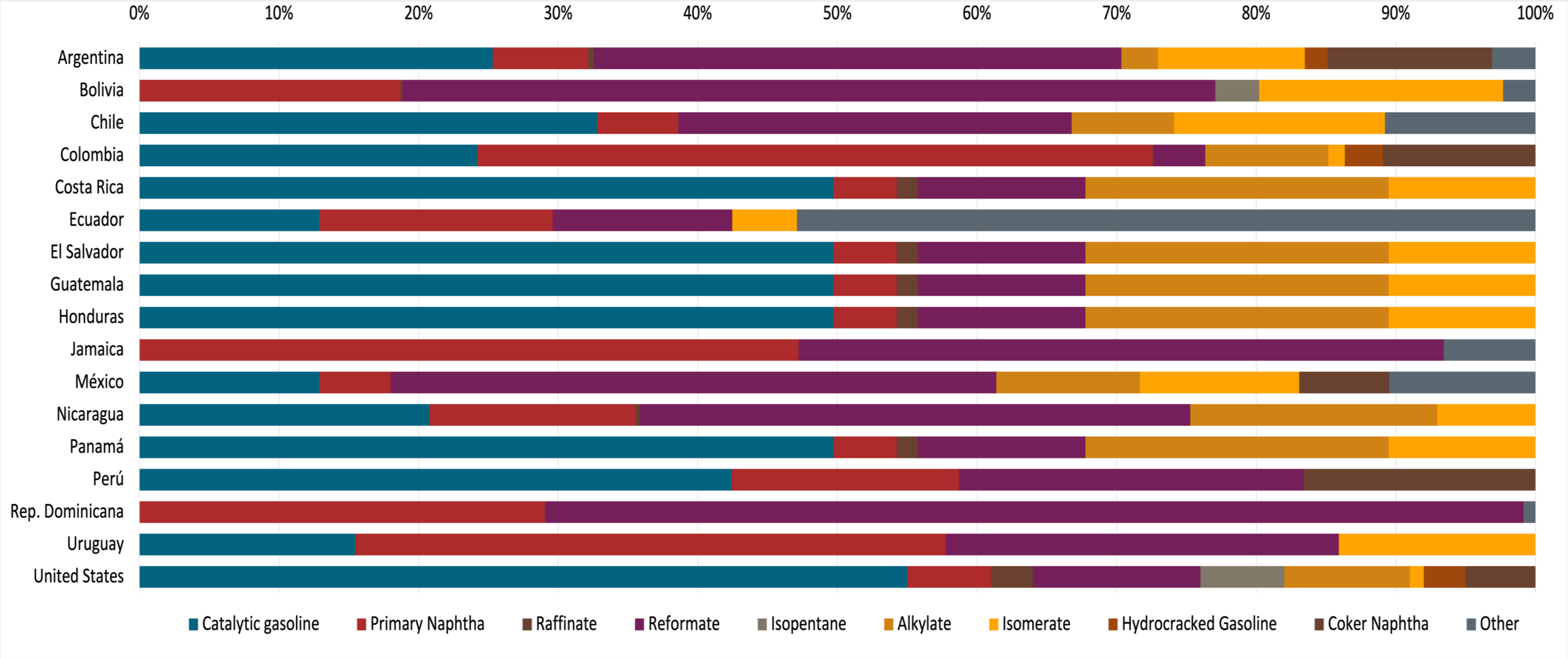
Gasoline Quality in Colombia

Name	Resolution 40103 de 2021				EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	07/01/2023				2017			
Applicability	Whole country	Whole country	Whole country	Whole country	All countries			
Selected Grade	Gasoline Regular	Gasoline Extra	Gasoline Regular E10	Gasoline Extra E10	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1,0 %v/v	< 2,0 %v/v	< 0,9 %v/v	< 1,8 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 28 %v/v	< 35 %v/v	< 25 %v/v	< 31,5 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	-	-	-	-	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	-	-	-	-	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 84	> 93	> 89	> 97	> 95	> 95	> 98	> 98
MON	-	-	-	-	> 85	> 88	> 85	> 88
AKI	> 81	> 91	> 84	> 94				
Sulfur Content	< 50 mg/kg	< 50 mg/kg	< 50 mg/kg	< 50 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content			< 3,7 %m/m	< 3,7 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	<> 9,5 - 10,5 %v/v	<> 9,5 - 10,5 %v/v	<> 9,5 - 10,5 %v/v	<> 9,5 - 10,5 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	<> 65 kPa	<> 65 kPa	<> 65 kPa	<> 65 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)								
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Source: Ministerio de Energía y Minas, 2021

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



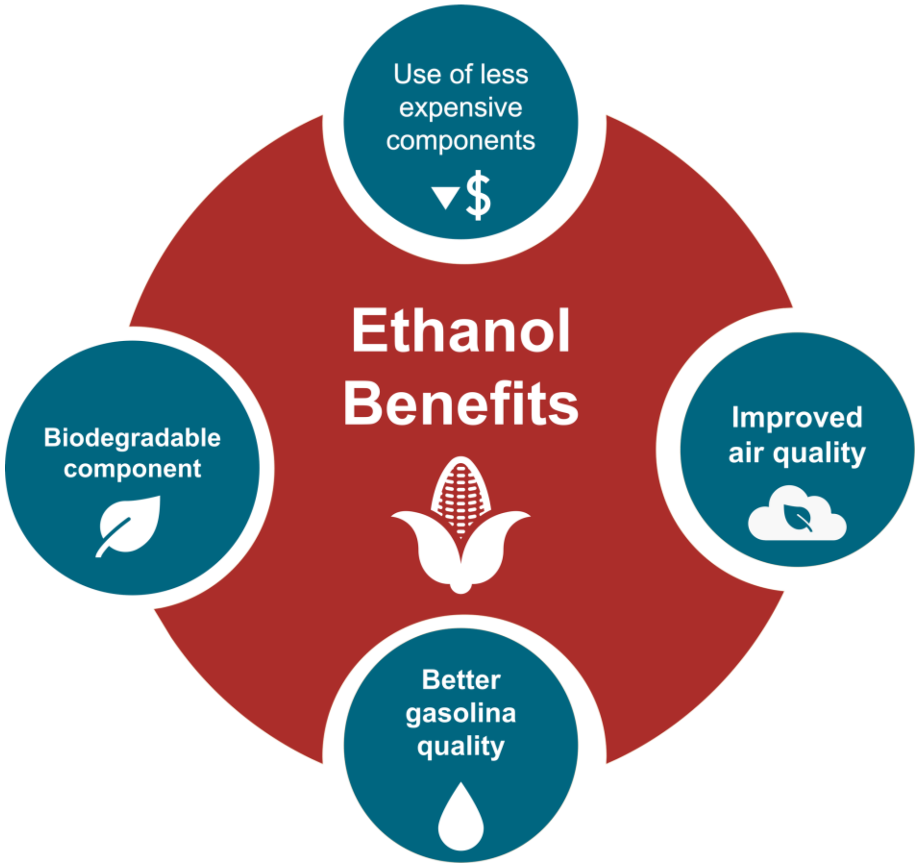
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

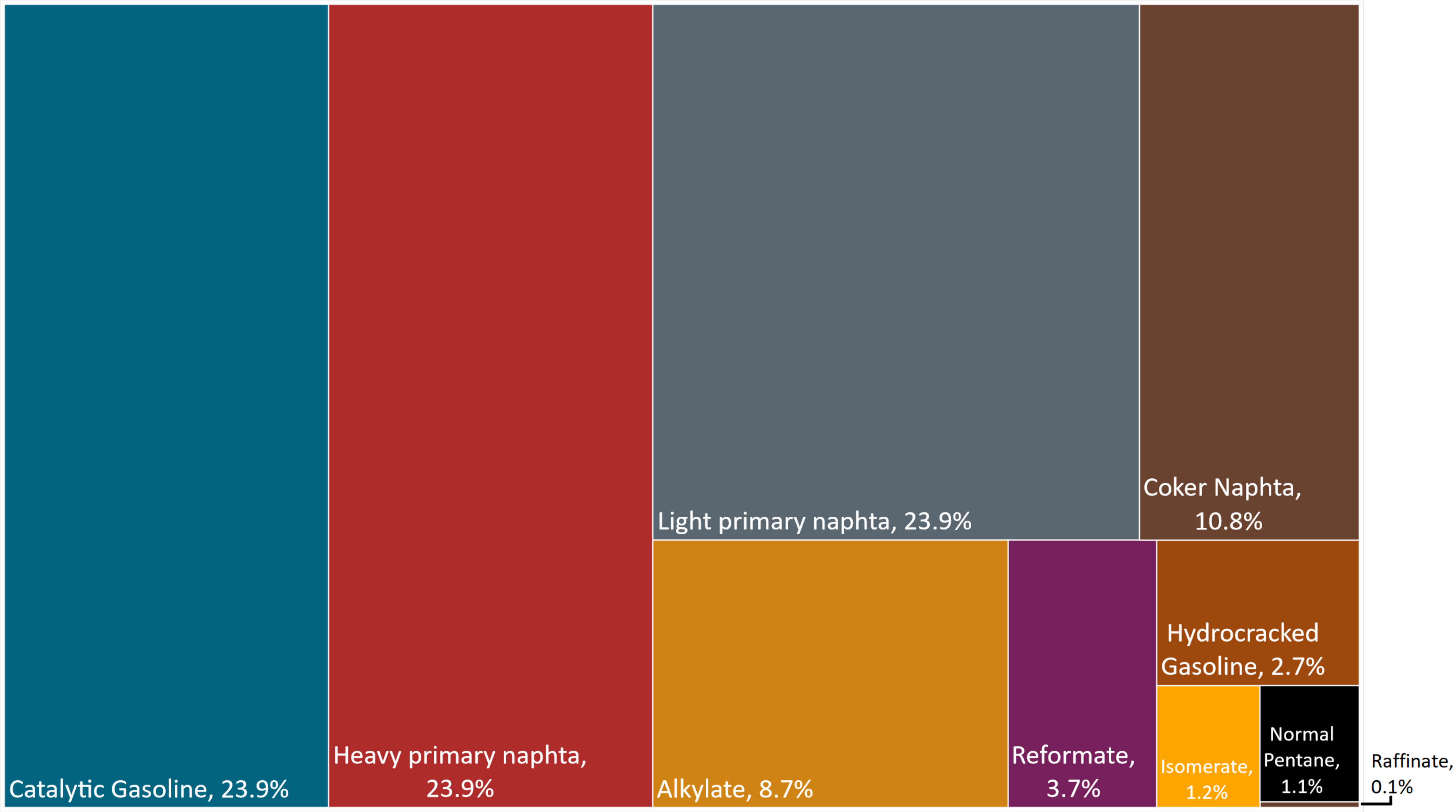
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices for 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

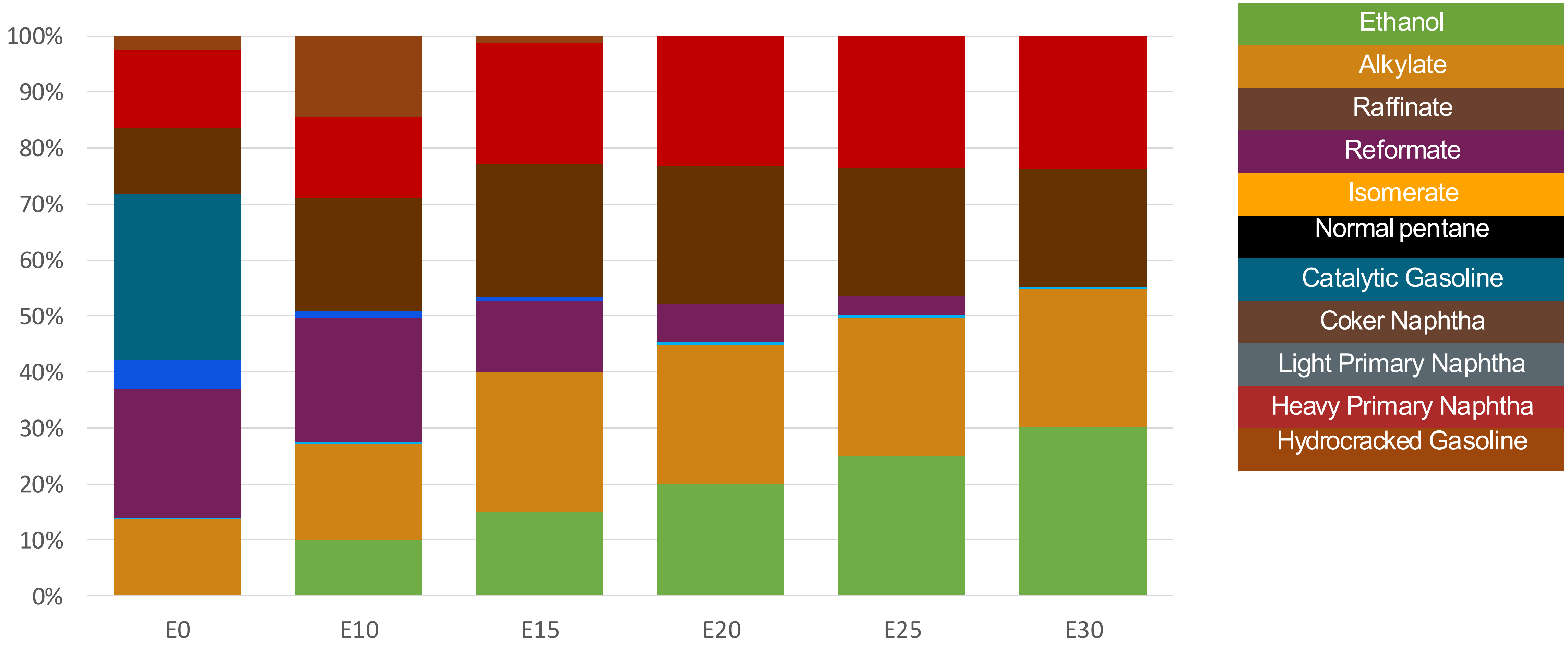


Available Blending Components



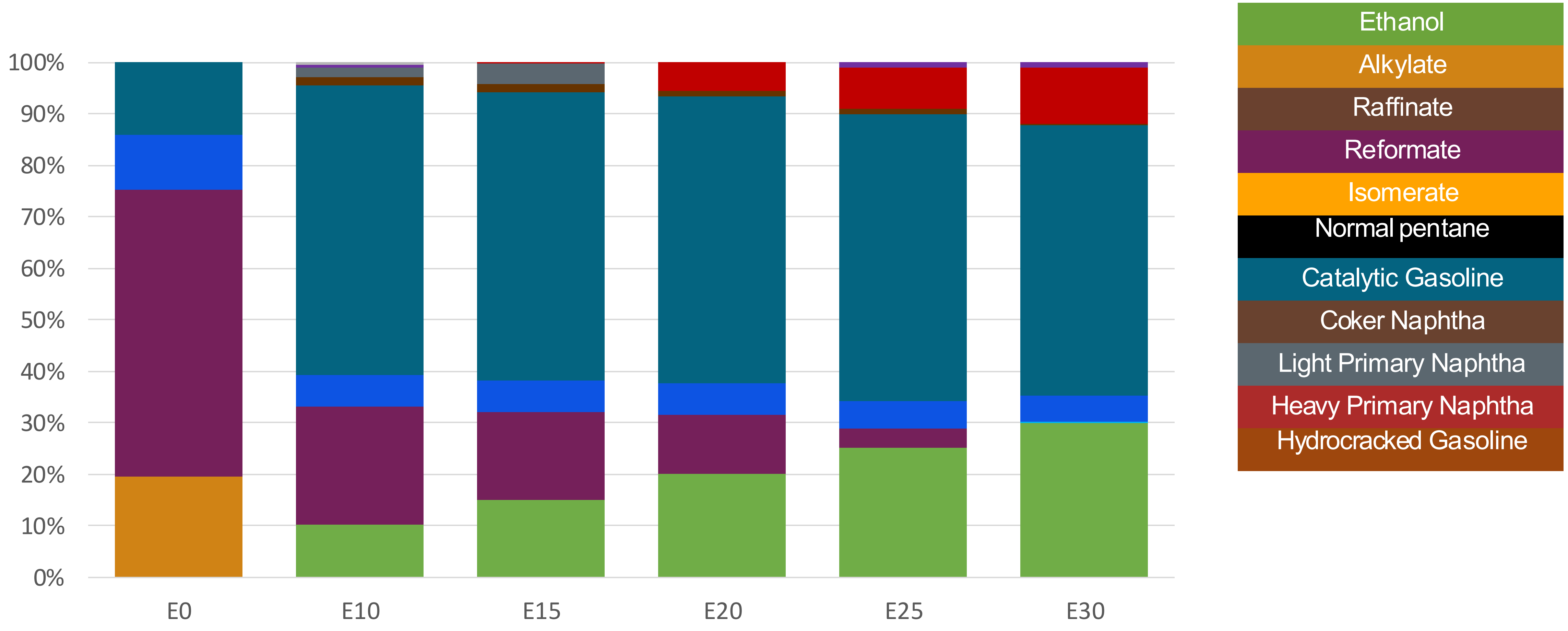
Source: Faro90

Colombia – Regular Gasoline – Constant Octane



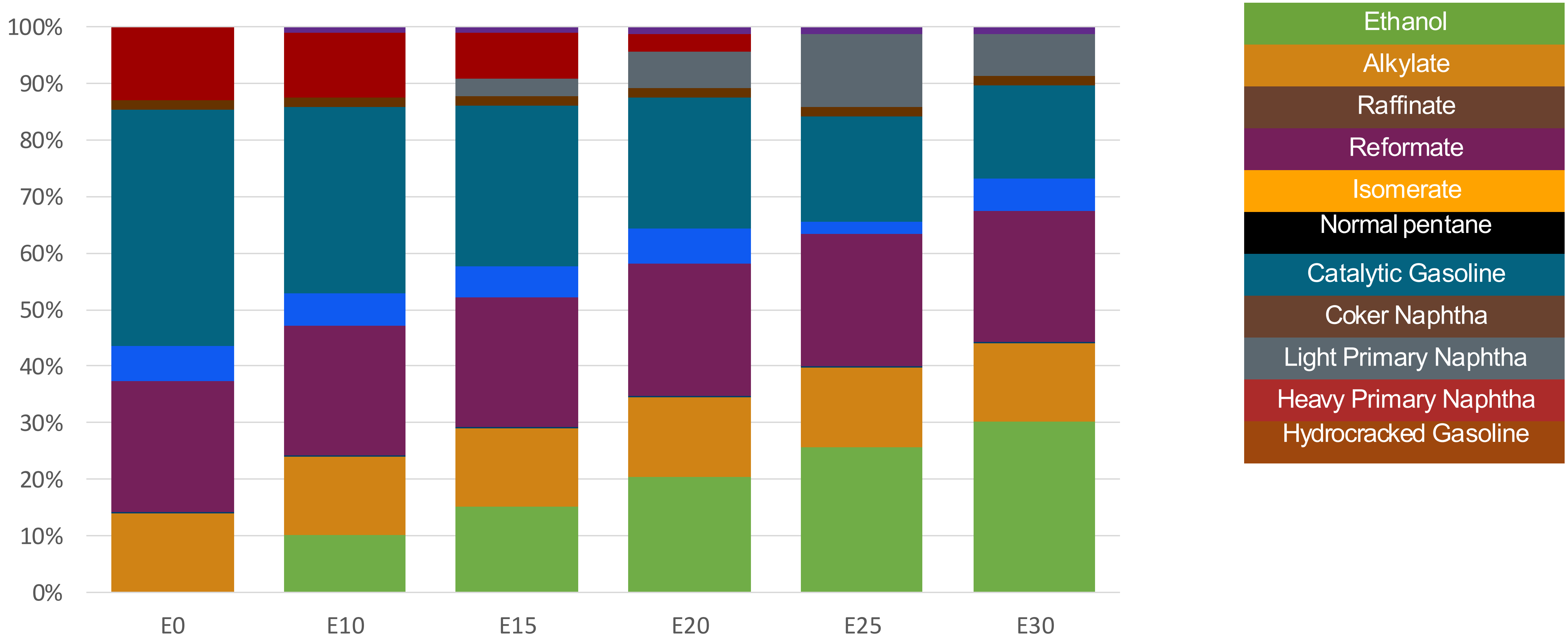
Octane (RON)	89.0	89.0	89.0	89.3	90.3	91.3
Price (US$/gal)	2.260	2.133	2.044	1.971	1.935	1.900

Colombia – Extra Gasoline – Constant Octane



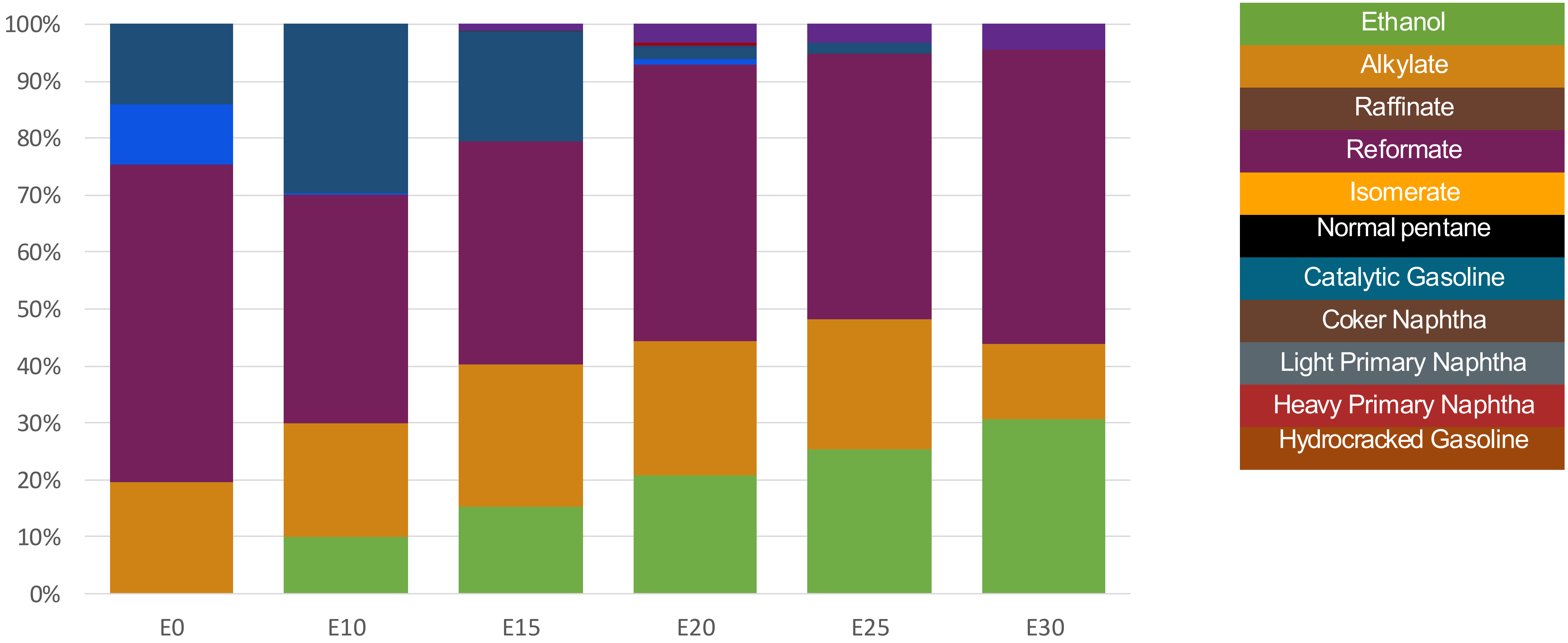
Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.0
Price (US$/gal)	2.660	2.451	2.352	2.264	2.185	2.100

Colombia – Regular Gasoline – Increased Octane



Octane (RON)	90.1	93.5	95.2	97.2	97.9	100.7
Price (US$/gal)	2.328	2.328	2.328	2.328	2.328	2.328

Colombia – Extra Gasoline – Increased Octane



Octane (RON)	97.0	98.2	99.6	101.4	103.0	105.1
Price (US$/gal)	$ 2.660	$ 2.660	$ 2.660	$ 2.660	$ 2.660	$ 2.660

Vehicle Emission Impact for Ethanol Gasoline Blending

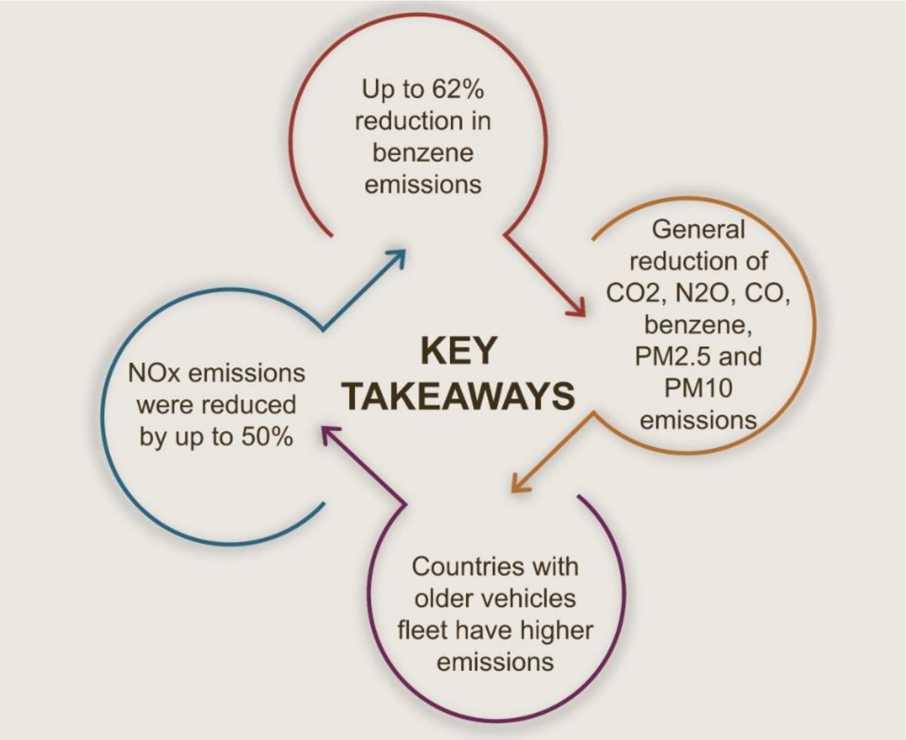
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



**Source: Bureau of transportation statistics.*

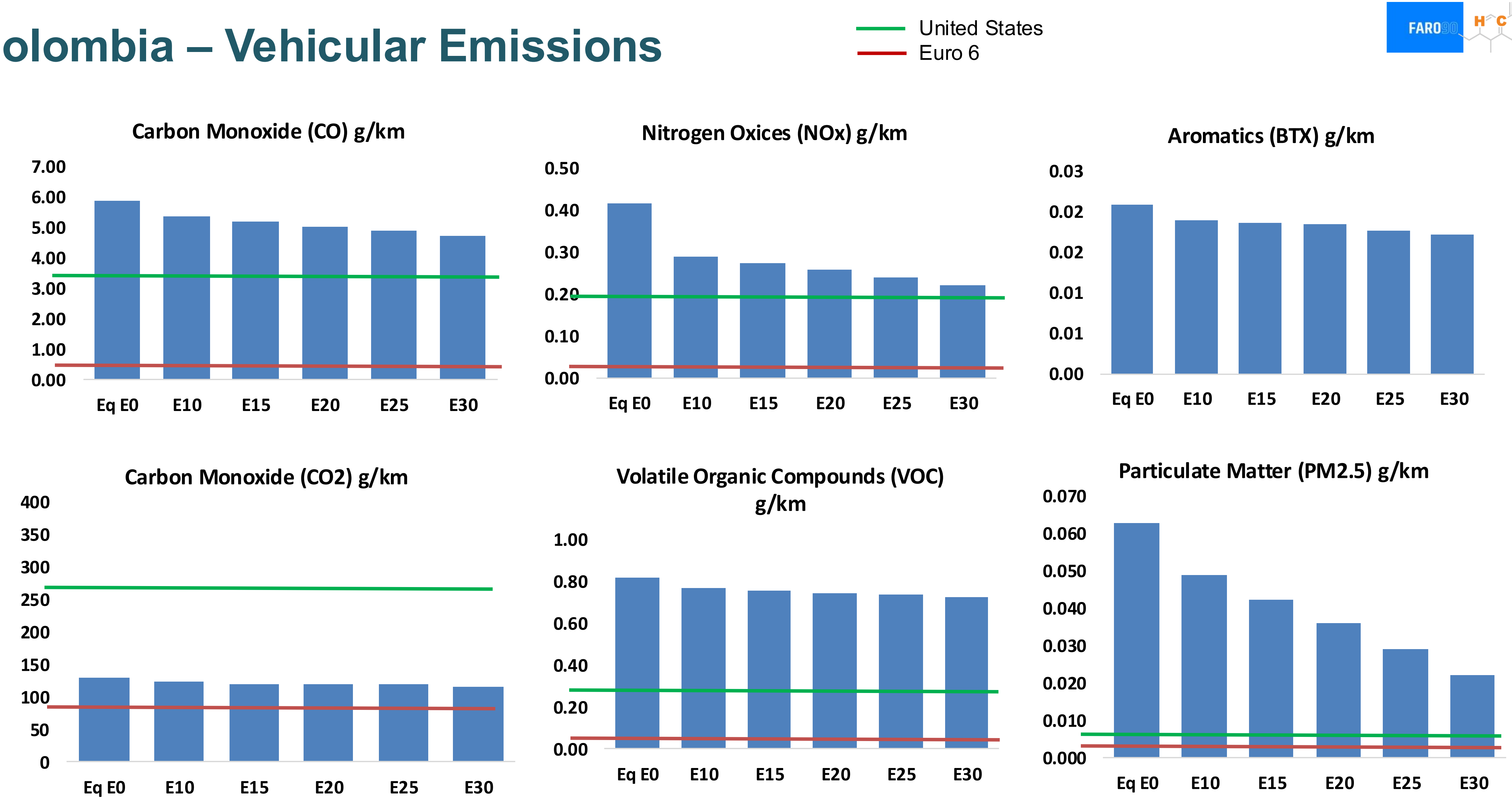
Gasoline Vehicle Fleet - Colombia

Tipo	Motor	Escape	Recuperator	Age	Number of vehicles
Light	Multi FI-Pt	Euro VI	PCV Tank	>4years, >80 mkm	386,511
		Euro V		4-8 years, 80-160 mkm	999,675
				<8 years, < 160 mkm	469,176
		Euro IV		<8 years, < 160 mkm	853,423
		Euro III		<8 years, < 160 mkm	1,011,893
		3-Ways		<8 years, < 160 mkm	2,487,481
Medium	Multi FI-Pt	Euro VI	PCV Tank	>4years, >80 mkm	300,891
		Euro V		4-8 years, 80-160 mkm	305,827
				<8 years, < 160 mkm	200,914
		Euro IV		<8 years, < 160 mkm	223,590
		Euro III		<8 years, < 160 mkm	201,232
		3-Ways		<8 years, < 160 mkm	60,439
Small Engines	FI 4-cycles	Euro V	PCV	>4years, >80 mkm	2,122,342
				4-8 years, 80-160 mkm	1,268,237
		Euro IV		4-8 years, 80-160 mkm	1,665,139
				<8 years, < 160 mkm	574,481
		Euro III		<8 years, < 160 mkm	5,852,593
		Euro II		<8 years, < 160 mkm	543,536
		none		<8 years, < 160 mkm	326,463
Heavy	FI	Euro VI	PCV	>4years, >80 mkm	54,048
		Euro V		4-8 years, 80-160 mkm	79,046
				<8 years, < 160 mkm	45,268
		Euro IV		<8 years, < 160 mkm	44,747
		Euro III		<8 years, < 160 mkm	28,167
		3-Ways		<8 years, < 160 mkm	14,108
			Total	**19,453,333**	

Vehicular Fleet: **19,453,333**
Average Age: **12.8 years**
Motorcycles Fleet: **62%**

Source: RUNT, Andemos, 2024, Faro90

Colombia – Vehicular Emissions



Source: Faro90

Colombia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,917,590	2,786,040	2,654,489	2,564,785	2,482,844	2,422,350	2,332,749	-9%	-15%
CO2	63,944,216	62,299,937	60,747,006	59,525,671	58,921,307	58,658,193	57,576,953	-5%	-8%
VOC	405,347	392,539	379,731	372,817	367,507	363,922	357,381	-6%	-9%
NOx	205,307	153,980	143,715	135,451	127,742	119,242	110,250	-30%	-38%
SOx	766	708	651	602	555	504	450	-15%	-28%
NH3	26,060	25,802	25,544	25,665	25,954	26,157	26,326	-2%	0%
N2O	1,518	1,510	1,504	1,511	1,542	1,559	1,577	-1%	2%
CH4	103,662	103,662	103,662	105,217	107,497	109,467	111,333	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	3,476	3,346	3,215	3,135	3,066	3,017	2,938	-8%	-12%
Acetaldehyde	9,260	10,628	15,594	23,747	32,353	36,970	43,684	68%	249%
Formaldehyde	37,231	36,197	42,196	49,547	51,908	57,424	62,512	13%	39%
Benzene	10,295	9,893	9,374	9,226	9,149	8,750	8,467	-9%	-11%
PM2.5	5,193	4,611	4,029	3,499	2,971	2,411	1,827	-22%	-43%
PM10	25,831	22,936	20,041	17,404	14,779	11,993	9,088	-22%	-43%

Source: Faro90

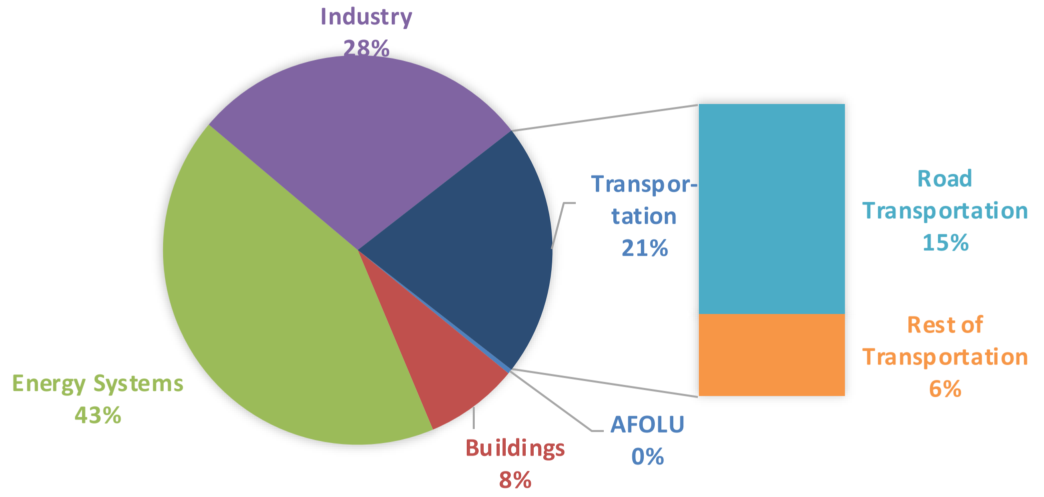
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

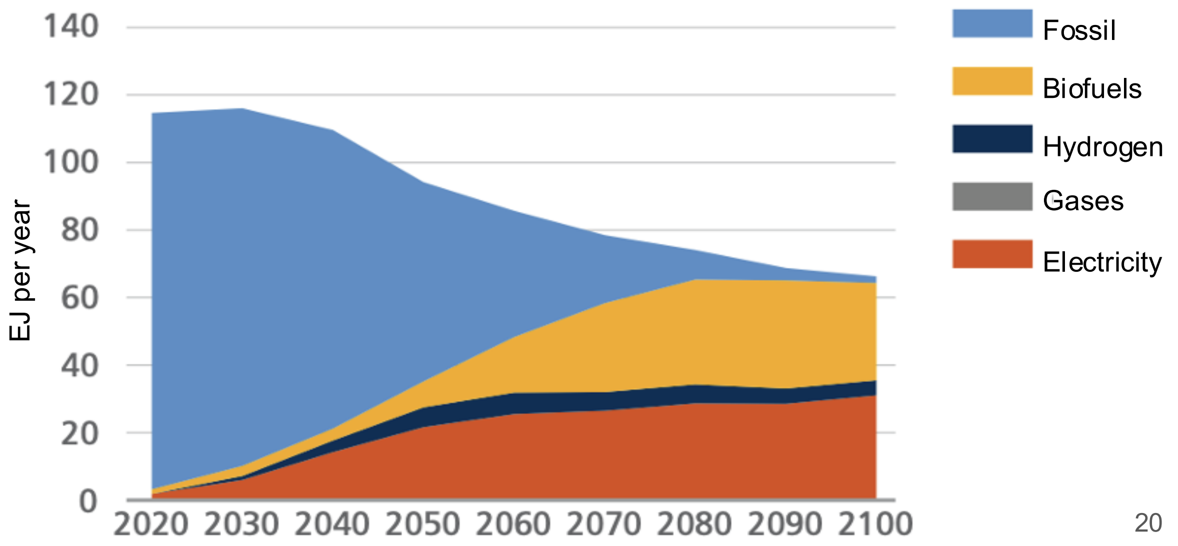
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

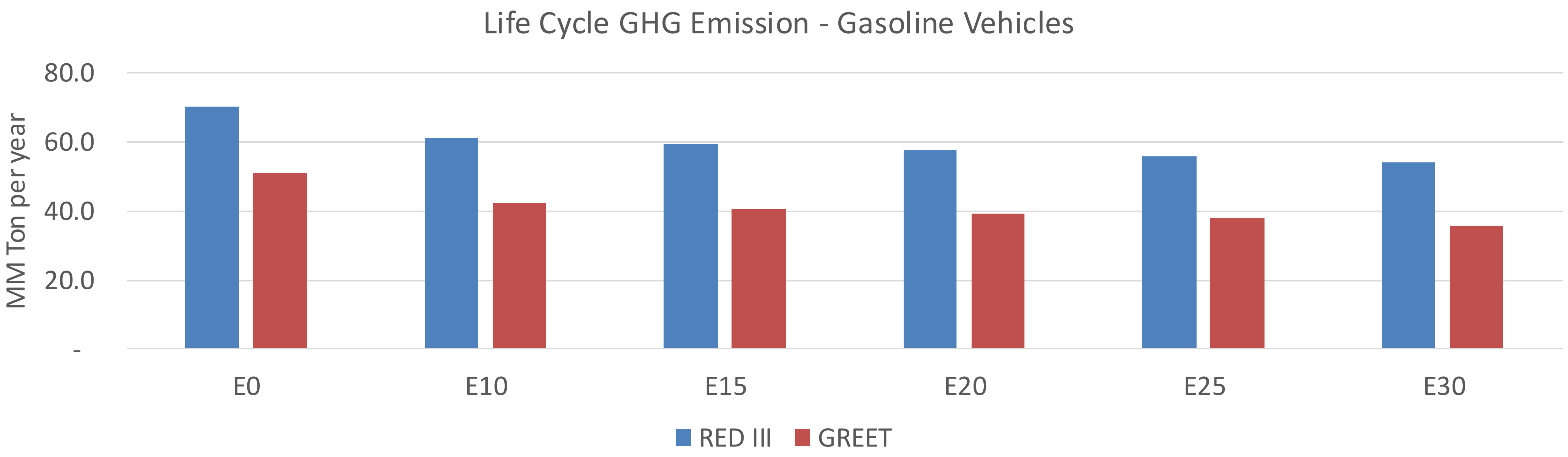
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Colombia – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	217.6
2035 Target (MMT/year)	125.5
Delta	92.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	17.0%	20.4%	23.3%	26.1%	29.7%
RED II/III	0.0%	13.0%	15.8%	18.2%	20.5%	23.3%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	9.4%	11.3%	12.9%	14.5%	16.5%
RED II/III	0.0%	9.9%	12.1%	13.9%	15.7%	17.8%

Source: Greet, RED II/III, climateactiontracker.org, F90